

The quiver associated to the stack of certain perfect chain complexes

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1 Introduction

2 Stack $\mathcal{X}$

Let $\mathcal{X}$ be the 2-stack :

$$\{\text{Commutative } \mathbb{C} \text{ algebra}\} \rightarrow \{2\text{-groupoid of two term perfect chain complexes with euler characteristics zero}\}.$$

Let $R_n = \mathbb{C}[x_{ij} | 1 \leq i, j \leq n]$. Consider the map

$$f_n : M_n(\mathbb{C}) = \text{Spec}(R_n) \rightarrow \mathcal{X}$$

corresponding to the chain complex

$$0 \rightarrow R_n \xrightarrow{\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nn} \end{pmatrix}} R_n \rightarrow 0.$$

For a commutative $\mathbb{C}$ algebra A , the map $f_n(A) : \text{Spec}(R_n)(A) = \text{Hom}(R_n, A) \rightarrow \mathcal{X}(A)$ is a 2-groupoid with

1. The objects in $f_n(A)$ are 2-term chain complexes of the form:

$$\phi \mapsto 0 \rightarrow A^n \xrightarrow{\begin{pmatrix} \phi(x_{11}) & \cdots & \phi(x_{1n}) \\ \vdots & \ddots & \vdots \\ \phi(x_{n1}) & \cdots & \phi(x_{nn}) \end{pmatrix}} A^n \rightarrow 0$$

where $\phi \in \text{Hom}(R_n, A)$ gives the R_n module structure to A .

2. The 1-morphisms are given by quasi isomorphisms between these chain complexes.
3. The 2-morphisms are given by chain homotopies between the quasi isomorphisms.

One can see that $f_n(A)$ forms a subcategory of $\mathcal{X}(A)$. We will show that the maps $\{f_n\}_{n \in \mathbb{N}}$ gives a covering of $\mathcal{X}$.

For any scheme Y and map g in the following diagram,

$$\begin{array}{ccc} M_n(\mathbb{C}) \times_{\mathcal{X}} Y & \xrightarrow{g'} & M_n(\mathbb{C}) \\ \downarrow f'_n & & \downarrow f_n \\ Y & \xrightarrow{g} & \mathcal{X} \end{array}$$

our goal is to show f'_n is smooth and surjective. It is enough to show assuming $Y = \text{Spec}(A)$ an arbitrary affine scheme where A is a finitely generated.

$$\begin{array}{ccccc} M_n(\mathbb{C}) \times_{\mathcal{X}} Y & \xrightarrow{h'} & M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) & \xrightarrow{g'} & M_n(\mathbb{C}) \\ \downarrow f'_n = h^* f''_n & & \downarrow f''_n & & \downarrow f_n \\ Y = \text{Spec}(A) & \xrightarrow{h} & M_m(\mathbb{C}) & \xrightarrow{f_m} & \mathcal{X} \\ & \searrow g & & & \end{array}$$

We start by factoring the map g via some map h and f_m . If we show f''_n (given above in the diagram) is smooth and surjective, then we get $h^* f''_n = f'_n$ smooth and surjective.

2.1 Understanding the fiber product

For commutative $\mathbb{C}$ algebra A , the fiber product $(M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A)$ is a 1-groupoid with

1. The objects are commutative diagrams

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & \downarrow a_0 & & \downarrow a_1 & & \\ 0 & \longrightarrow & A^m & \xrightarrow{t_1} & A^m & \longrightarrow & 0 \end{array}$$

such that (a_0, a_1) is a quasi isomorphism. We will denote such commutative diagram using a tuple $(t_0, t_1 : a_0, a_1)$.

2. The 1-morphisms are given by chain homotopies between quasi isomorphisms i.e.

$$\text{Mor}((t_0, t_1 : a_0, a_1), (t_0, t_1 : a'_0, a'_1)) = \{f : A^n \rightarrow A^m \mid a_0 - a'_0 = f t_0 \text{ and } a_1 - a'_1 = t_1 f\}.$$

This can be depicted as:

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & \downarrow a'_0 & \swarrow f & \downarrow a'_1 & & \\ & & A^m & \xrightarrow{t_1} & A^m & \longrightarrow & 0 \end{array}$$

Moreover if $(t_0, t_1) \neq (t'_0, t'_1)$, then

$$\text{Mor}((t_0, t_1 : a_0, a_1), (t'_0, t'_1 : a'_0, a'_1)) = \emptyset.$$

Lemma 1: For commutative $\mathbb{C}$ algebra A , consider the following diagram (not necessarily commutative)

$$\begin{array}{ccccccc} 0 & \longrightarrow & A^n & \xrightarrow{t_0} & A^n & \longrightarrow & 0 \\ & & \downarrow a_0 & & \downarrow a_1 & & \\ 0 & \longrightarrow & A^m & \xrightarrow{t_1} & A^m & \longrightarrow & 0. \end{array}$$

The diagram lies in $(M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A)$ if and only if the maps satisfy the following:

1. $t_1 a_0 - a_1 t_0 = [t_1 \ a_1] [a_0 \ t_0]^T = 0$
2. $[a_0 \ t_0]^T$ is split injective i.e. there exists a left inverse.
3. $[t_1 \ a_1]$ is surjective

Proof. A chain map (a_0, a_1) is a quasi isomorphism between (t_0, t_1) is equivalent to proving the mapping cone being acyclic i.e. the following is a short exact sequence

$$0 \longrightarrow A^n \xrightarrow{[a_0 \ t_0]^T} A^{n+m} \xrightarrow{[t_1 \ a_1]} A^m \longrightarrow 0$$

which implies $[t_1 \ a_1] [a_0 \ t_0]^T = 0$, $[a_0 \ t_0]^T$ is injective, $[t_1 \ a_1]$ is surjective and $\text{im}[a_0 \ t_0]^T = \ker [t_1 \ a_1]$.

Since objects in the above short exact sequence are free A -modules, the above short exact sequence splits and $[a_0 \ t_0]^T$ being injective gives split injectivity which proves implication in one direction.

Suppose $[t_1 \ a_1] [a_0 \ t_0]^T = 0$, $[a_0 \ t_0]^T$ is split injective and $[t_1 \ a_1]$ is surjective. Since $[t_1 \ a_1]$ is surjective map between free modules, we get

$$A^{n+m} \cong \ker [t_1 \ a_1] \oplus A^m.$$

Using $[t_1 \ a_1] [a_0 \ t_0]^T = 0$, we have

$$\text{im}[a_0 \ t_0]^T \subseteq \ker [t_1 \ a_1]$$

and $[a_0 \ t_0]^T$ being split injectivity gives

$$\ker [t_1 \ a_1] = K \oplus \text{im}[a_0 \ t_0]^T \cong K \oplus A^n$$

for some A -module K ,

Using above congruences,

$$A^{n+m} \cong K \oplus A^n \oplus A^m.$$

For any prime ideal p in A , rank of K_p is 0 which implies the A_p module $K_p = 0$. Thus, $K = 0$ and $\ker [t_1 \ a_1] = \text{im}[a_0 \ t_0]^T$ which proves our result. $\square$

Let $Z_{n,m}$ be the functor of sets given in the following way: For commutative $\mathbb{C}$ algebra A ,

$$Z_{n,m}(A) = \text{The set of objects of category } (M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A)$$

Any set can be regarded as 1-groupoid by considering elements as objects and setting automorphism group to identity and no morphisms between distinct elements.

Thus, $Z_{n,m}(A)$ is a 1-groupoid with

1. The objects are commutative diagrams

$$(t_0, t_1 : a_0, a_1)$$

i.e. (a_0, a_1) is a quasi isomorphism between $(t_0 \in M_n(A), t_1 \in M_m(A))$.

2. The 1-morphisms are given by

$$\text{Mor}((t_0, t_1 : a_0, a_1), (t'_0, t'_1 : a'_0, a'_1)) = \begin{cases} \{id\} & (t_0, t_1 : a_0, a_1) = (t'_0, t'_1 : a'_0, a'_1) \\ 0 & \text{else} \end{cases}$$

We are interested to know if the functor $Z_{n,m}$ is representable. We define some representable subfunctors of $Z_{n,m}$. For $I, J \subset \{1, 2, \dots, n+m\}$ such that $|I| = n$ and $|J| = m$, let

$$\begin{aligned} Z_{n,m,I,J} = \text{Spec } \mathbb{C}[(t_0, t_1, a_0, a_1, r_1, r_2) | [t_1 \ a_1] [a_0 \ t_0]^T = 0, \\ r_1 \det [a_0 \ t_0]_I^T = 1 \text{ and} \\ r_2 \det [t_1 \ a_1]_J = 1]. \end{aligned}$$

Note that

$$\mathbb{C}[(t_0, t_1, a_0, a_1, r_1, r_2) | [t_1 \ a_1] [a_0 \ t_0]^T = 0, r_1 \det [a_0 \ t_0]_I^T = 1 \text{ and } r_2 \det [t_1 \ a_1]_J = 1]$$

is a finitely generated ring with variables

$$\begin{aligned} t_0 &= ((t_0)_{ij}) & 1 \leq i, j \leq n \\ t_1 &= ((t_1)_{ij}) & 1 \leq i, j \leq m \\ a_0 &= ((a_0)_{ij}) & 1 \leq i \leq m, 1 \leq j \leq m \\ a_1 &= ((a_1)_{ij}) & 1 \leq i \leq m, 1 \leq j \leq n \end{aligned}$$

and, r_1 and r_2 are considered as single variables.

Lemma 2: The functor $Z_{n,m}$ is representable.

Proof. For a commutative $\mathbb{C}$ algebra A , observe that for a scheme $\text{Spec}(R) \xrightarrow{\phi} Z_{n,m}$, we have the following pullback diagram

$$\begin{array}{ccc} \phi^{-1}(D(\det [a_0 \ t_0]_I^T, \det [t_1 \ a_1]_J)) & \longrightarrow & Z_{n,m,I,J}(A) \\ \downarrow & & \downarrow \\ \text{Spec}(R)(A) & \xrightarrow{\phi} & Z_{n,m}(A) \end{array}$$

where $D(\det [a_0 \ t_0]_I^T, \det [t_1 \ a_1]_J)$ is the open subset of $Z_{n,m}(A)$ such that $\det [a_0 \ t_0]_I^T, \det [t_1 \ a_1]_J \in A^*$. This shows that $Z_{n,m,I,J}$ are open representable subfunctors of $Z_{n,m}$.

Using Theorem 8.9 from [GW20], it is enough to construct a surjective map from $\bigcup_{I,J} Z_{n,m,I,J} \hookrightarrow Z_{n,m}$. Moreover [Sta26, Tag 06D4], it is enough to check the surjectivity condition over fields.

For a field k , consider an object $(t_0 \in k^{n^2}, t_1 \in k^{m^2} : a_0 \in k^{nm}, a_1 \in k^{nm}) \in Z_{n,m}(k)$. Using 1, the matrix $[a_0 \ t_0]^T : k^n \rightarrow k^{n+m}$ is split injective and the matrix $[t_1 \ a_1] : k^{n+m} \rightarrow k^m$ is surjective. This implies rank of $[a_0 \ t_0]^T$ is n and rank of $[t_1 \ a_1]$ is m . Thus, there exists a maximal minor which is invertible that shows $\bigcup_{I,J} Z_{n,m,I,J}(k) \hookrightarrow Z_{n,m}(k)$ is surjective. $\square$

Without loss of generality, let's fix $m \leq n$. For $I, J \subset \{1, 2, \dots, n+m\}$ such that $|I| = n$ and $|J| = m$, define the maps from the ring map associated to the representable functors

$$\begin{aligned} T_0^{I,J} : Z_{n,m,I,J} &\rightarrow M_n(\mathbb{C}) \\ (t_0, t_1, a_0, a_1, r_0, r_1) &\rightarrow t_0 \end{aligned}$$

and

$$\begin{aligned} T_1^{I,J} : Z_{n,m,I,J} &\rightarrow M_m(\mathbb{C}) \\ (t_0, t_1, a_0, a_1) &\rightarrow t_1 \end{aligned}$$

Since $\bigcup_{I,J} Z_{n,m,I,J} \rightarrow Z_{n,m}$ is surjective and

$$\begin{aligned} T_0^{I,J}|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}} &= T_0^{I',J'}|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}} \\ T_1^{I,J}|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}} &= T_1^{I',J'}|_{Z_{n,m,I,J} \cap Z_{n,m,I',J'}}, \end{aligned}$$

we get the maps $T_0 : Z_{n,m} \rightarrow M_n\mathbb{C}$ and $T_1 : Z_{n,m} \rightarrow M_m\mathbb{C}$ such that both the triangles in the following diagram commutes

$$\begin{array}{ccc} \bigcup_{I,J} Z_{n,m,I,J} & \xrightarrow{\quad} & Z_{n,m} \\ & \searrow \cup_{I,J} T_0^{I,J} & \downarrow T_0 \\ & & M_n\mathbb{C} \end{array} \quad \begin{array}{ccc} \bigcup_{I,J} Z_{n,m,I,J} & \xrightarrow{\quad} & Z_{n,m} \\ & \searrow \cup_{I,J} T_1^{I,J} & \downarrow T_1 \\ & & M_m\mathbb{C} \end{array}$$

Moreover, by universal property of fiber product, there exists a unique map Q such that the triangles in the following diagram commutes

$$\begin{array}{ccccc} Z_{n,m} & & & & \\ & \searrow Q & \searrow T_1 & & \\ & & M_n\mathbb{C} \times_{\mathcal{X}} M_m\mathbb{C} & \xrightarrow{\quad} & M_n\mathbb{C} \\ & \searrow T_0 & \downarrow & & \downarrow \\ & & M_m\mathbb{C} & \xrightarrow{\quad} & \mathcal{X} \end{array}$$

Lemma 3: The map T_0 is smooth and surjective.

Proof. For commutative $\mathbb{C}$ algebra A , by consturction of $Z_{n,m}$, it is clear that T_0 is surjective. Since $Z_{n,m}$ is covered by $Z_{n,m,I,J}$, we will show that $T_0^{I,J} : Z_{n,m,I,J}(A) \rightarrow M_m(\mathbb{C})(A)$ is smooth. We know that

$$\begin{aligned} Z_{n,m,I,J}(A) = \text{Spec } \mathbb{C}[(t_0, t_1 : a_0, a_1, r_1, r_2) \mid t_0 \in M_n(A), t_1 \in M_m(A) \\ [t_1 \ a_1] [a_0 \ t_0]^T = 0, \\ r_1 \det [a_0 \ t_0]_I^T = 1 \text{ and} \\ r_2 \det [t_1 \ a_1]_J = 1]. \end{aligned}$$

Set

$$\begin{aligned} t_0 &= \{t_{ij}\}_{1 \leq i, j \leq n}, \\ t_1 &= \{t'_{ij}\}_{1 \leq i, j \leq m}, \\ a_0 &= \{a_{ij}\}_{1 \leq i \leq m, 1 \leq j \leq n}, \\ a_1 &= \{a'_{ij}\}_{1 \leq i \leq m, 1 \leq j \leq n}, \\ d_1 &= \det [a_0 \ t_0]_I^T \\ d_2 &= \det [t_1 \ a_1]_J. \end{aligned}$$

Then it must satisfy the following equations:

$$\begin{aligned} e_{ij} &:= \sum_{k=1}^m t'_{ik} a_{kj} - \sum_{k=1}^n a'_{ik} t_{kj} = 0 & \forall 1 \leq i \leq m, 1 \leq j \leq n, \\ z_1 &:= r_1 d_1 = 1 & \text{and} \\ z_2 &:= r_2 d_2 = 1. \end{aligned}$$

Observe that

$$\begin{aligned} \frac{\partial}{\partial t'_{ij}}(e_{i'j'}) &= \delta_{ii'} a_{jj'} \\ \frac{\partial}{\partial a'_{ij}}(e_{i'j'}) &= -\delta_{ii'} t_{jj'} \end{aligned}$$

Then the jacobian matrix J of the map $T_0^{I,J}$ is a $(m^2 + 2mn + 2) \times (mn + 2)$ matrix of the form

$$J = \begin{matrix} \frac{\partial}{\partial a_{ij}} & \frac{\partial}{\partial t'_{1j}} & \vdots & \frac{\partial}{\partial t'_{mj}} & \vdots & \frac{\partial}{\partial a'_{1j}} & \vdots & \frac{\partial}{\partial a'_{mj}} & \frac{\partial}{\partial r_1} & \frac{\partial}{\partial r_2} \end{matrix} \begin{pmatrix} (e_{1j})_{1 \leq j \leq n} & \cdots & (e_{mj})_{1 \leq j \leq n} & z_1 & z_2 \\ * & * & * & * & * \\ a_0 & 0 & 0 & * & * \\ 0 & \ddots & 0 & * & * \\ 0 & 0 & a_0 & * & * \\ -t_0 & 0 & 0 & * & * \\ 0 & \ddots & 0 & * & * \\ 0 & 0 & -t_0 & * & * \\ 0 & 0 & 0 & d_1 & 0 \\ 0 & 0 & 0 & 0 & d_2 \end{pmatrix}$$

Since d_1, d_2 are invertible and $[a_0 \ t_0]$ is of rank n , the matrix J has rank $mn + 2$ i.e. full rank. This implies $T_0^{I,J}$ is smooth and hence the result. $\square$

Consider an additive group $M_{m,n} = \text{Spec } \mathbb{C}[(x_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}]$. Then for any commutative $\mathbb{C}$ algebra A , there is an action of $M_{m,n}(A)$ on $Z(A)$ given in the following way:

$$X \cdot (t_0, t_1 : a_0, a_1) = (t_0, t_1 : a_0 + z t_0, a_1 + t_1 z)$$

for all $z \in M_{m,n}(A)$ and $(t_0, t_1 : a_0, a_1) \in Z_{n,m}(A)$.

Proposition 4: For an additive group G_a , any principal G -bundle over an affine scheme is trivial.

Proof. Isomorphism classes of principal G_a -bundles over a scheme X are classified by the first cohomology group $H_{\text{fppf}}^1(X, \mathbb{G}_a)$.

Since $\mathbb{G}_a$ is a smooth group scheme, its cohomology in the fppf or étale topology coincides with its cohomology in the Zariski topology. Furthermore, for any quasi-coherent sheaf $\mathcal{F}$ on a scheme X , we have

$$H_{\text{fppf}}^1(X, \mathbb{G}_a) \cong H_{\text{ét}}^1(X, \mathbb{G}_a) \cong H_{\text{Zar}}^1(X, \mathcal{O}_X)$$

If X is an affine scheme, then the cohomology groups :

$$H^1(X, \mathcal{O}_X) = 0$$

$\square$

Lemma 5: The quotient stack $[Z_{n,m}/M_{m,n}]$ is isomorphic to $M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C})$

Proof. For a commutative $\mathbb{C}$ algebra A , the objects in $[Z_{n,m}/M_{m,n}](A)$ are

$$\{(p, f) \mid p : P \rightarrow \text{Spec}(A) \text{ is a } M_{m,n}\text{-bundle, } f : P \rightarrow Z_{n,m} \text{ is } M_{m,n}\text{-equivariant}\}.$$

Since $M_{m,n}$ is an additive group that tells us any principal $M_{m,n}$ -bundle over an affine variety is trivial by 4. This implies that the objects in $[Z_{n,m}/M_{m,n}](A)$ are

$$\{p : M_{m,n} \times \text{Spec}(A) \rightarrow Z_{n,m} \mid p \text{ is an } M_{m,n}\text{-equivariant map}\}.$$

Using equivariance of $M_{m,n}$, any such map p can be determined from $p|_{id_{M_{m,n}} \times \text{Spec}(A)} : id_{M_{m,n}} \times \text{Spec}(A) = \text{Spec}(A) \rightarrow Z_{n,m}$ which is an object in $Z_{n,m}(A)$. Since the objects in $Z_{n,m}(A)$ are same as the objects of $(M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A)$, we get the following 1-1 correspondence:

$$\begin{aligned} [Z_{n,m}/M_{m,n}](A) &\longleftrightarrow (M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A) \\ p &\longmapsto p|_{id_{M_{m,n}} \times \text{Spec}(A)}. \end{aligned}$$

Any morphism between p_1 and p_2 is an $M_{m,n}$ -equivariant isomorphism θ which makes the following diagram commute

$$\begin{array}{ccccc} & & M_{m,n} \times \text{Spec}(A) & & \\ & \swarrow \pi & \downarrow \theta & \searrow p_1 & \\ \text{Spec}(A) & & & & Z_{n,m} \\ & \nwarrow \pi & \downarrow & \swarrow p_2 & \\ & & M_{m,n} \times \text{Spec}(A) & & \end{array}$$

where π_2 is the second coordinate projection map. Since the left triangle commutes, we can write $\theta(z, x) = (\theta_1(z, x), x)$ for some $M_{m,n}$ -equivariant map $\theta_1 : M_{m,n} \times \text{Spec}(A) \rightarrow M_{m,n}$. Since θ_1 is an $M_{m,n}$ -equivariant map, it is parametrized by the maps $\theta_1(1, x) : id_{M_{m,n}} \times \text{Spec}(A) = \text{Spec}(A) \rightarrow M_{m,n}$ i.e. an element of $M_{m,n}(A)$. The map θ_1 must also satisfy

$$p_2(\theta_1(1, x), x) = p_1(1, x)$$

for all $x \in \text{Spec}(A)$. Suppose p_1 and p_2 are $M_{m,n}$ -equivariant map that corresponds to element $(t_0, t_1 : a_0, a_1)$ and $(t'_0, t'_1 : a'_0, a'_1)$ in $(M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}))(A)$ respectively i.e.

$$\begin{aligned} p_1|_{id_{M_{m,n}} \times \text{Spec}(A)} &= (t_0, t_1, a_0, a_1) \\ p_2|_{id_{M_{m,n}} \times \text{Spec}(A)} &= (t'_0, t'_1, a'_0, a'_1). \end{aligned}$$

Then

$$\begin{aligned} Mor(p_1, p_2) &= \{z \in M_{m,n}(A) \mid p_2(z, x) = p_1(1, x) \ \forall x \in \text{Spec}(A)\} \\ &= \{z \in M_{m,n}(A) \mid z \cdot p_2(1, x) = p_1(1, x) \ \forall x \in \text{Spec}(A)\} \\ &= \{z \in M_{m,n}(A) \mid z \cdot (t'_0, t'_1, a'_0, a'_1) = (t_0, t_1, a_0, a_1)\} \\ &= \{z \in M_{m,n}(A) \mid (t'_0, t'_1, a'_0 + zt'_0, a'_1 + t'_1 z) = (t_0, t_1, a_0, a_1)\} \end{aligned}$$

This implies that $Mor(p_1, p_2) = \emptyset$, if $(t_0, t_1) \neq (t'_0, t'_1)$. For $(t_0, t_1) = (t'_0, t'_1)$,

$$Mor(p_1, p_2) = \{z \in M_{m,n}(A) \mid a_0 - a'_0 = zt_0 \text{ and } a_1 - a'_1 = t_1 z\}$$

Thus, $[Z_{n,m}/M_{m,n}](A)$ and $M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C})(A)$ are equivalent categories for all A which proves the lemma. $\square$

Corollary 6: The map $M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is smooth and surjective.

Proof. Since $Z_{n,m} \rightarrow [Z_{nm}/M_{m,n}] \cong M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C})$ is smooth, surjective and $Z_{n,m} \rightarrow M_m(\mathbb{C})$ is smooth and surjective, we get $M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ is smooth and surjective. $\square$

This shows that $f_n : M_n\mathbb{C} \rightarrow \mathcal{X}$'s forms a covering.

3 Understanding $Z_{n,m}$

Lets work with the case when $A = \mathbb{C}$. Suppose t_0 and t_1 are matrices of different corank, then it is easy to show that they are not quasi isomorphic to each other. Let $t_0 = \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix}$, $t_1 = \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix}$, $a_0 = \{x_{ij}\}_{1 \leq i \leq m, 1 \leq j \leq n}$ and $a_1 = \{y_{ij}\}_{1 \leq i \leq m, 1 \leq j \leq n}$. Then we can find all the quasi isomorphism (a_0, a_1) between t_0 and t_1 using the following equations:

$$\begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix} \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix} = \begin{bmatrix} y_{11} & \cdots & y_{1n} \\ \vdots & \ddots & \vdots \\ y_{m1} & \cdots & y_{mn} \end{bmatrix} \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix},$$

$$\begin{aligned} a_0|_{\ker(t)} : \ker(t_0) &\rightarrow \ker(t_1) \\ [a_1] : \mathbb{C}^n / \text{Im}(t_0) &\rightarrow \mathbb{C}^n / \text{Im}(t_1) \end{aligned}$$

are isomorphisms. Solving for x_{ij} and y_{ij} , we get a_0 and a_1 are of the form

$$\begin{aligned} a_0 &= \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} \text{ and} \\ a_1 &= \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix} \end{aligned}$$

where $X_k, Y_k \in GL_k(\mathbb{C})$, $X \in M_{k,n-k}(\mathbb{C})$, $Y \in M_{m-k,k}(\mathbb{C})$ and $Z \in M_{m-k,n-k}(\mathbb{C})$. Since any square matrix can be row and column reduced to matrix of the form $\begin{bmatrix} 0_k & 0 \\ 0 & I_{k'} \end{bmatrix}$, we define

$$Z_{n,m} := \bigcup_{k=0}^{\min\{n,m\}} Z_{n,m,k}$$

where

$$\begin{aligned} Z_{n,m,k} &:= \{(Q \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix} P^{-1}, R \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix} S^{-1}, S \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} P^{-1}, R \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix} Q^{-1}) | \\ &P, Q \in GL_n(\mathbb{C}), R, S \in GL_m(\mathbb{C}), X_k, Y_k \in GL_k(\mathbb{C}), \\ &Z \in M_{m-k,n-k}(\mathbb{C}), X \in M_{k,n-k}(\mathbb{C}) \text{ and } Y \in M_{m-k,k}(\mathbb{C})\}. \end{aligned}$$

Let $G := (GL_n(\mathbb{C}) \times GL_n(\mathbb{C}) \times GL_m(\mathbb{C}) \times GL_m(\mathbb{C})) \ltimes M_{m,n}(\mathbb{C})$ where the group operation is given by

$$(P', Q', R', S', x') \cdot (P, Q, R, S, x) = (P'P, Q'Q, R'R, S'S, x + S^{-1}x'Q).$$

Observe that the group G acts on $Z_{n,m}$ as follows:

$$(P, Q, R, S, x) \cdot (t_0, t_1, a_0, a_1) = (Qt_0P^{-1}, Rt_1S^{-1}, S(a_0 + xt_0)P^{-1}, R(a_1 + t_1x)Q^{-1})$$

For $g_1 = (P, Q, R, S, x), g_2 = (P', Q', R', S', x') \in G$ and $z = (t_0, t_1, a_0, a_1) \in Z_{n,m}$,

$$\begin{aligned} g_2 \cdot (g_1 \cdot z) &= (P', Q', R', S', x') \cdot (Qt_0P^{-1}, Rt_1S^{-1}, S(a_0 + xt_0)P^{-1}, R(a_1 + t_1x)Q^{-1}) \\ &= (Q'Qt_0P^{-1}(P')^{-1}, R'Rt_1S^{-1}(S')^{-1}, S'(S(a_0 + xt_0)P^{-1} + x'Qt_0P^{-1})(P')^{-1}, \\ &\quad R'(R(a_1 + t_1x)Q^{-1} + Rt_1S^{-1}x')(Q')^{-1}) \\ &= ((Q'Q)t_0(P'P)^{-1}, (R'R)t_1(S'S)^{-1}, (S'S)(a_0 + (x + S^{-1}x'Q)t_0)(P'P)^{-1}, \\ &\quad (R'R)(a_1 + t_1(x + S^{-1}x'Q))(Q'Q)^{-1}) \\ &= (P'P, Q'Q, R'R, S'S, x + S^{-1}x'Q) \cdot (t_0, t_1, a_0, a_1) \\ &= (g_2 \cdot g_1) \cdot z \end{aligned}$$

Lemma 7: The G -orbits of $Z_{n,m}$ are precisely $Z_{n,m,k}$ for $0 \leq k \leq \min\{n, m\}$.

Proof. Let $z = (Q \begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix} P^{-1}, R \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix} S^{-1}, S \begin{bmatrix} X_k & X \\ 0 & Z \end{bmatrix} P^{-1}, R \begin{bmatrix} Y_k & 0 \\ Y & Z \end{bmatrix} Q^{-1})$ be an element of $Z_{n,m,k}$ for some $P, Q \in GL_n(\mathbb{C}), R, S \in GL_m(\mathbb{C}), X_k, Y_k \in GL_k(\mathbb{C}), Z \in M_{m-k, n-k}(\mathbb{C}), X \in M_{k, n-k}(\mathbb{C})$ and $Y \in M_{m-k, k}(\mathbb{C})$. Set

$$g = \left(P \begin{bmatrix} X_k^{-1} & 0 \\ 0 & I_{n-k} \end{bmatrix}, Q, R \begin{bmatrix} Y_k & 0 \\ 0 & I_{m-k} \end{bmatrix}, S, \begin{bmatrix} 0 & X \\ Y & Z \end{bmatrix} \right)$$

and

$$z_0 = \left(\begin{bmatrix} 0_k & 0 \\ 0 & I_{n-k} \end{bmatrix}, \begin{bmatrix} 0_k & 0 \\ 0 & I_{m-k} \end{bmatrix}, \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} \right).$$

Then $g \cdot z_0 = z$. This shows that $Z_{n,m,k}$ lies inside an orbit of G . Since G will not change the corank of t_0 or t_1 , the subsets $Z_{n,m,k}$'s lie in different orbits which proves the lemma. $\square$

For $n, m \in \mathbb{N}$, let's denote the map $\pi_1^{n,m} : M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ and $\pi_2^{n,m} : M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) \rightarrow M_m(\mathbb{C})$ such that the following diagram commutes

$$\begin{array}{ccc} M_n(\mathbb{C}) \times_{\mathcal{X}} M_m(\mathbb{C}) & \xrightarrow{\pi_1^{n,m}} & M_n(\mathbb{C}) \\ \pi_2^{n,m} \downarrow & & \downarrow f_n \\ M_m(\mathbb{C}) & \xrightarrow{f_m} & \mathcal{X}. \end{array}$$

Lemma 8: For $n, m \in \mathbb{N}$ and $i = 1, 2$, the fibers of $\pi_i^{n,m}$ are connected.

Corollary 9: For $n, m \in \mathbb{N}$ and $n \leq m$, the functor $(\pi_1^{n,m})^*$ is fully faithful.

4 Perverse Sheaves

Let $\mathcal{X}_n := f_n(M_n(\mathbb{C}))$ be the open substack of $\mathcal{X}$. Since $f_n : M_n(\mathbb{C}) \rightarrow \mathcal{X}_n$ is smooth and surjective, we can define perverse sheaf on $\mathcal{X}_n$ in the following manner:

Definition 10: A G -equivariant perverse sheaf of $\mathcal{X}_n$ is a tuple (F, ϕ) where F is a $GL_n(\mathbb{C}) \times GL_n(\mathbb{C})$ -equivariant perverse sheaf in $M_n(\mathbb{C})$ and ϕ is an isomorphism

$$\pi_1^{n,n} F \xrightarrow{\phi} \pi_2^{n,n} F$$

which satisfies the descent data defined in [Ach21] (definition 3.7.1).

Since f_n have connected fibers, we have $Perv_{GL_n(\mathbb{C}) \times GL_n(\mathbb{C})}(M_n(\mathbb{C})) \hookrightarrow Perv_G(\mathcal{X}_n)$ is fully faithful functor.

Theorem 11: For $n \in \mathbb{N}$,

$$Perv_G(\mathcal{X}_n) \cong Perv_{GL_n(\mathbb{C}) \times GL_n(\mathbb{C})}(M_n(\mathbb{C}))$$

Proof. It is enough to show the projective covers of IC_k in $Perv_{GL_n(\mathbb{C}) \times GL_n(\mathbb{C})}(M_n(\mathbb{C}))$ lies in $Perv_G(\mathcal{X}_n)$.

For a projective cover P , there exists an isomorphism $\pi_1^{n,n} P \cong \pi_2^{n,n} P$. Since the isomorphism is a scalar, the cocycle condition can be a scalar apart. We can normalize to set the cocycle condition equal and get the result.

For an arbitrary perverse sheaf, using the following diagram

$$\begin{array}{ccccccc} F & \xrightarrow{\pi_2^{n,n}} & \pi_2^{n,n}(F) & \xrightarrow{\sim} & \pi_1^{n,n}(F) & \xleftarrow{\pi_1^{n,n}} & F \\ \uparrow & & \uparrow \text{coker} & & \uparrow \text{coker} & & \uparrow \\ P^\bullet & \xrightarrow{\pi_2^{n,n}} & \mathcal{P}^\bullet & \xrightarrow{\sim} & \mathcal{P}^\bullet & \xleftarrow{\pi_1^{n,n}} & P^\bullet \end{array}$$

we can produce an isomorphism. Hence the embedding is also essentially surjective and hence equivalent as categories. □

Definition 12: We define $Perv_G(\mathcal{X}) := \varprojlim_i Perv_G(\mathcal{X}_i)$.

Theorem 13: $Perv_G(\mathcal{X}) \cong Rep(AA_\infty)$

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